

Tyler Zeng

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Projects

NEXUS: A Novel Deafblind Communication Device [5x National Awards]

Python, C, Embedded Systems 🔗

- Designed a wearable accessibility device that converts conversations into speaker-attributed transcripts, which can be accessed through accessible formats for deafblind individuals.
- Constructed using **Raspberry Pi 5** and **STM32**, with a custom 6-microphone array and a modular interface that supports several methods of accessible output (i.e. braille, enlarged text, and haptics).
- Integrated a novel, dynamic speaker identification framework that surpasses state-of-the-art methods for real-time, far-field environments, fusing spatial localization (**TDOA**) and vocal feature extraction (**Pyannote**).

Fluctus: An Accessible Digital Hearing Aid [National Finalist]

Python, SciPy, NumPy 🔗

- Prototyped a fully digital hearing aid application with real-time adaptive amplification, featuring a **10-band equalizer**, noise suppression (**DeepFilterNet**), vocal restoration (**VoiceFixer**), and tinnitus support.
- Iterated multiple software configurations to remove reliance on hardware, lowering barriers to hearing treatment.

Longitudinal Multi-Omics Modeling for Early Risk Stratification in IBD

Python, PyTorch, Scikit-learn 🔗

- Designed a novel, proof of concept, dual-encoder deep learning model in **PyTorch** to predict inflammatory bowel disease progression, iterating on data pipelines across **130 patients and 1,600+ samples**.
- Built a **Gaussian mixture model** to surface **three** molecularly distinct Crohn's disease patient subgroups, enabling early treatment stratification and precision medicine applications.
- Identified a high-risk subgroup representing **10%** of patients with significantly elevated molecular inflammation, demonstrating the model's potential for actionable clinical insight.

SIGMA: Sign Identification with Guided Machine Assistance

Python, OpenCV, YOLOv8, Figma 🔗

- Built a multilingual AI road sign recognition system, leveraging smartphone cameras and **YOLOv8** and a **custom dataset**, achieving **95%** real-time detection accuracy.
- Planned expansion into a fully integrated mobile application for real-time deployment.

Awards

- Science Fair** — Nationals: Youth Can Innovate Grand Award, Silver Medal, Best in Category, Canadian Acoustical Association Award, \$10,000+ in scholarships, 2x Finalist | Regionals: 2x Gold Medal, 2x Best in Category
- VEX Robotics** — Worlds Qualifier 2026, Provincial Champion, 4x Performance Awards, 6x Judged Awards

Experiences

Sunnybrook Research Institute | Incoming Engineering Intern

July 2026 - August 2026

- Contributing to the development and validation of ultrasound-based medical technologies through hardware integration, experimental testing, and engineering research.

Science Fair Club | Co-Founder & Co-President

Oct 2025 – Present

- Co-founded and scaled a **50+** member student club, overseeing an executive team across various operations.
- Designed and delivered workshops on research design, experimentation, and scientific presentation.

VEX Robotics Team 3004A | Design & Build Lead

Mar 2024 – Present

- Led the iterative design and construction of competition robots, coordinating team timelines and strategies.
- Mentored new members in mechanical design and **CAD**, ensuring technical consistency and quality across robots.
- Adapted match strategies in real time during tournament play, contributing to multiple championship placements.

Education

St. Robert Catholic High School — IB Diploma Programme

Grade 11 | 98% Average

Skills

Technologies: Python, C, Embedded Systems, Fusion 360, OnShape, Git